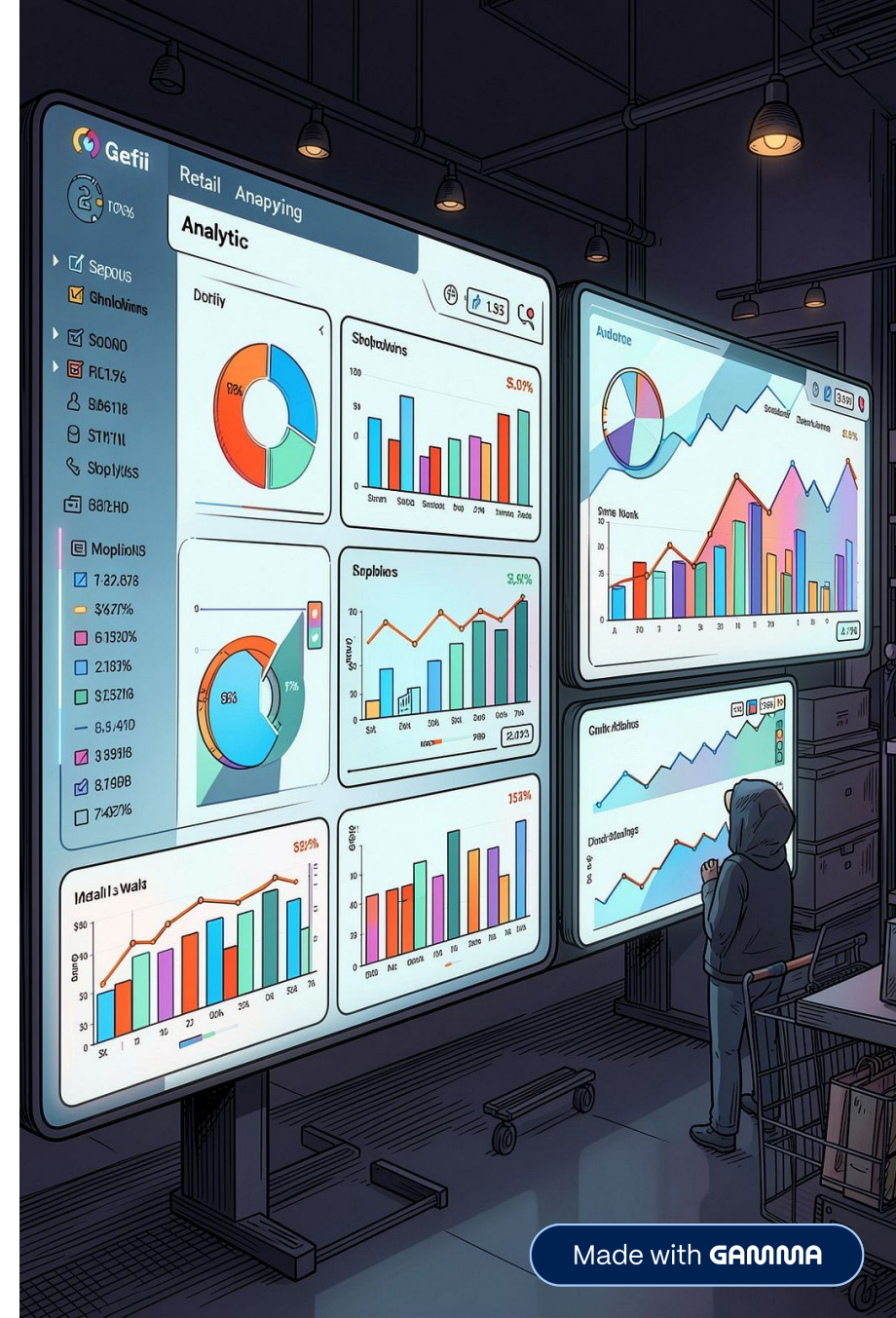


Customer Shopping Behavior Analysis

Uncovering spending patterns, segments, and preferences across **3,900 purchases**



Dataset at a Glance

3,900

Purchases

18

Columns

37

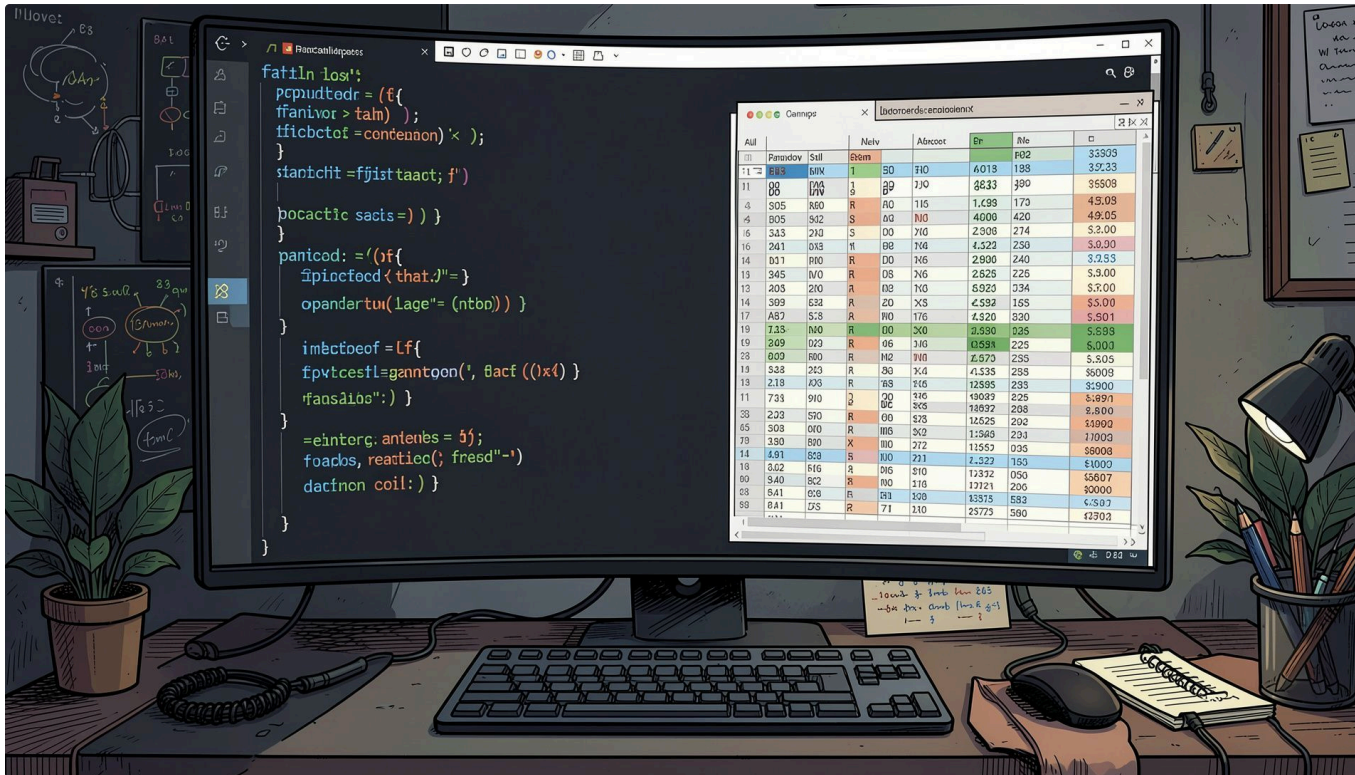
Missing Values

Review Rating only

50

Locations

Data Preparation in Python



→ Clean & Impute

Median fill for missing Review Ratings by category

→ Standardize

Renamed columns to snake_case

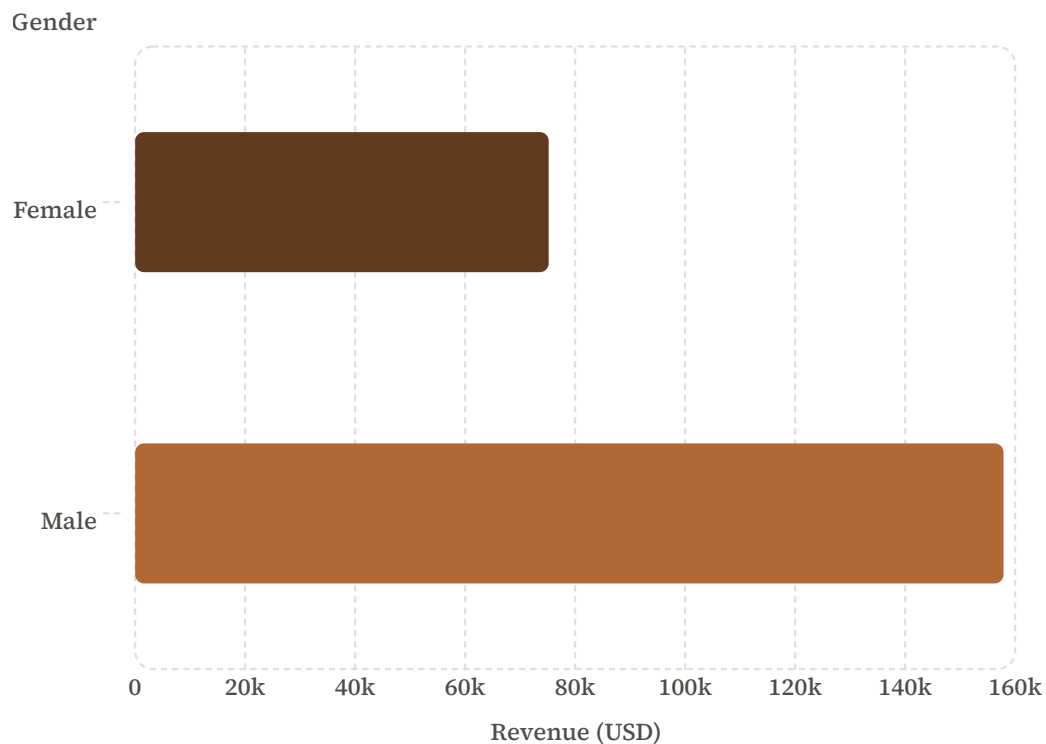
→ Engineer

Created **age_group** & **purchase_frequency_days**

→ Load to DB

Connected to PostgreSQL for SQL analysis

Revenue by Gender



Male customers drive 2× more revenue

Male: **\$157,890** · Female: **\$75,191**

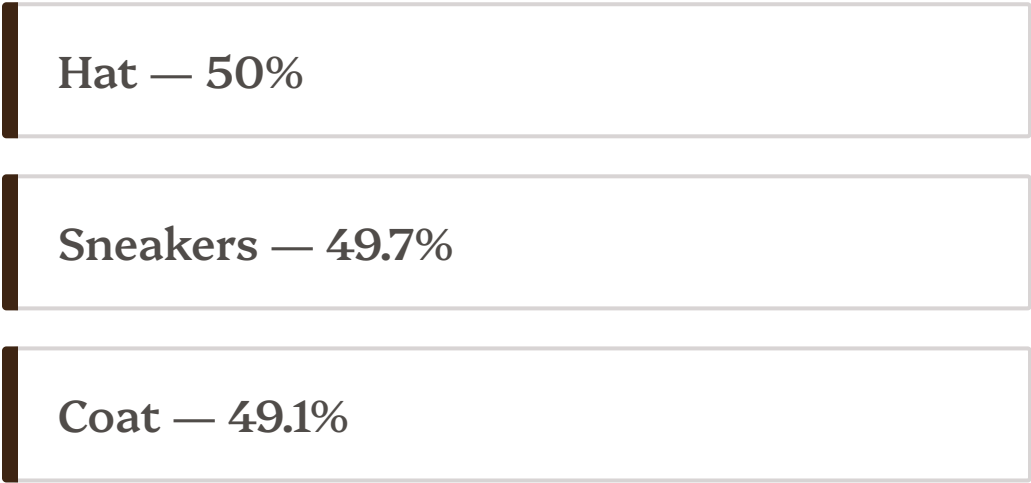
Male shoppers also account for **2,652 of 3,900** transactions
— over two-thirds of all purchases.

Product Insights

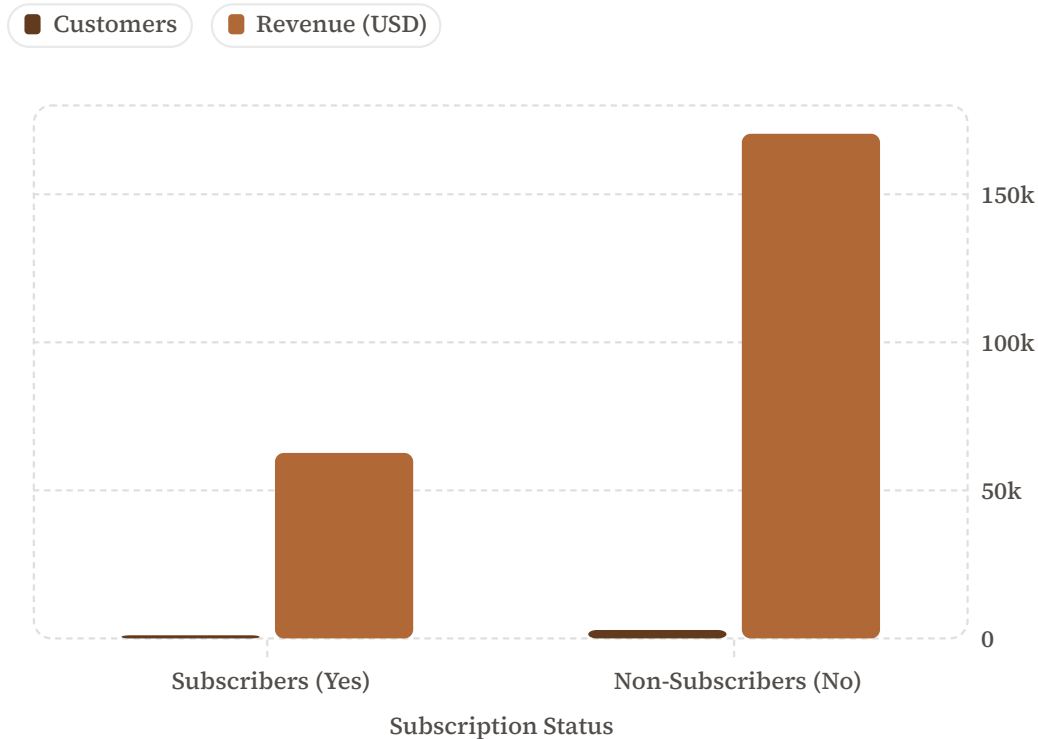
Top 5 Products by Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products



Subscribers vs. Non-Subscribers



Non-subscribers dominate volume

73% of customers are non-subscribers, generating **\$170K** in revenue.

Average spend is nearly identical: **\$59.49** (subscribers) vs. **\$59.87** (non-subscribers).

Of customers with **>5 purchases**, 958 subscribe vs. 2,518 who don't — repeat buyers are not converting to subscriptions.

Customer Segmentation

 Loyal

3,116 customers

 Returning

701 customers

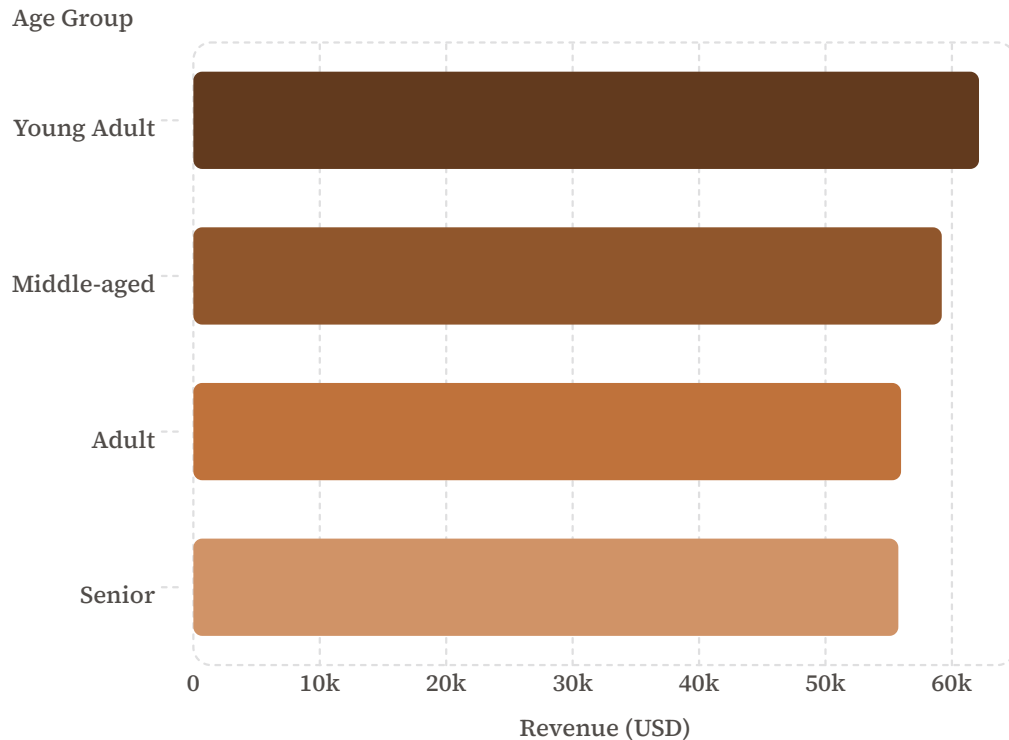
 New

83 customers

Over **80%** of the customer base is already in the Loyal segment — a strong retention foundation.



Revenue by Age Group



Young Adults lead revenue

Average age: **44 years** (range 18–70). Revenue is fairly distributed across age groups, with Young Adults slightly ahead at **\$62,143**.

Express shipping customers spend slightly more on average (**\$60.48**) vs. Standard (**\$58.46**).

Power BI Dashboard



Revenue by Category

Clothing leads at \$90K



Subscription Split

27% Yes · 73% No



Interactive Filters

Gender, Category, Shipping, Age

Business Recommendations

1

Boost Subscriptions

Promote exclusive subscriber benefits

2

Loyalty Programs

Reward repeat buyers → Loyal segment

3

Review Discount Policy

Balance sales lift vs. margin control

4

Product Positioning

Highlight top-rated & best-selling items

5

Targeted Marketing

Focus on high-revenue age groups & express-shipping users